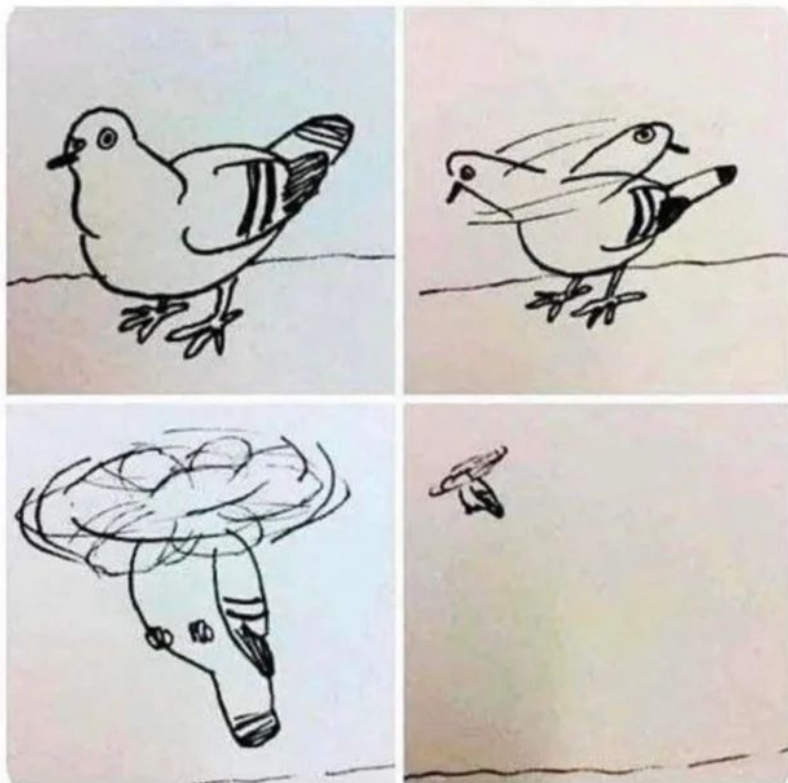
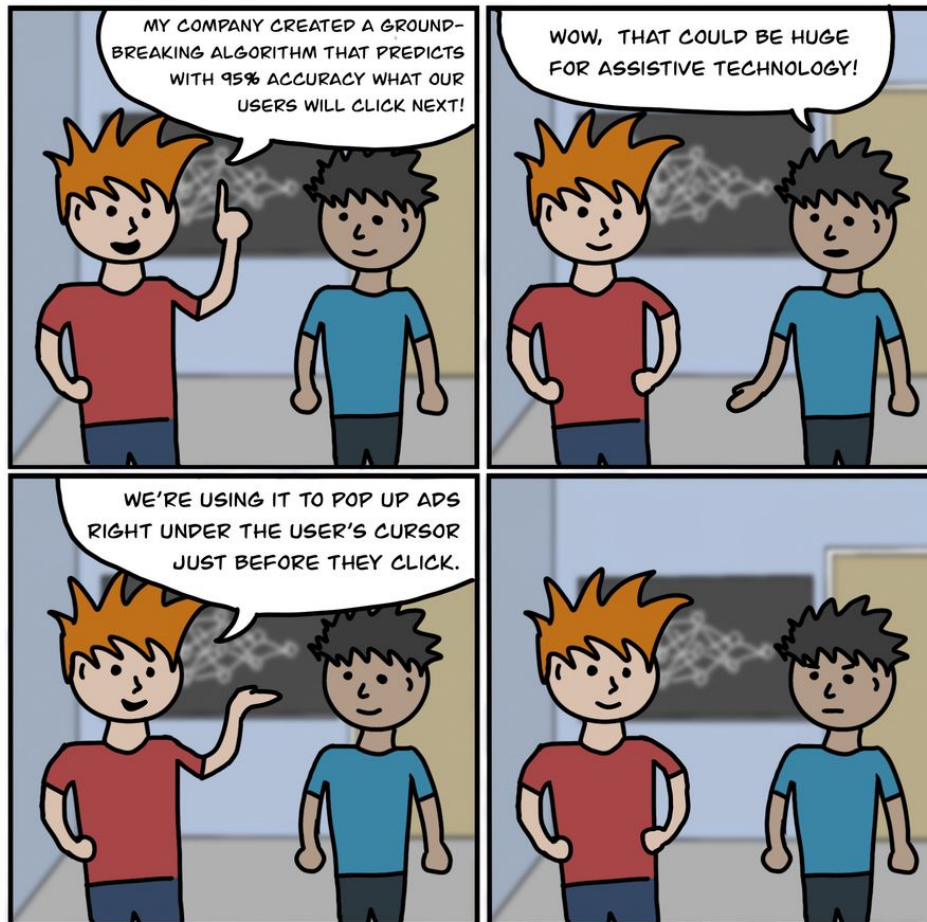


When your program
is a complete mess,
but it does its job



GROUNDBREAKING



Hey you! Let's chat.



Have you ever heard of the free and open-source operating system called Linux?

maybe



It was created in 1991 by Linus Torvalds. It is very popular now, as it is used on most servers and desktops.

Should I use it instead of windows?



You should use Linux instead of Windows.

[See more](#)

Type a message...





Freya Holmér

@FreyaHolmer



btw these large scary math symbols
are just for-loops

Summation
(capital sigma)

$$\sum_{n=0}^4 3n$$

```
sum = 0;  
for( n=0; n<=4; n++ )  
    sum += 3*n;
```

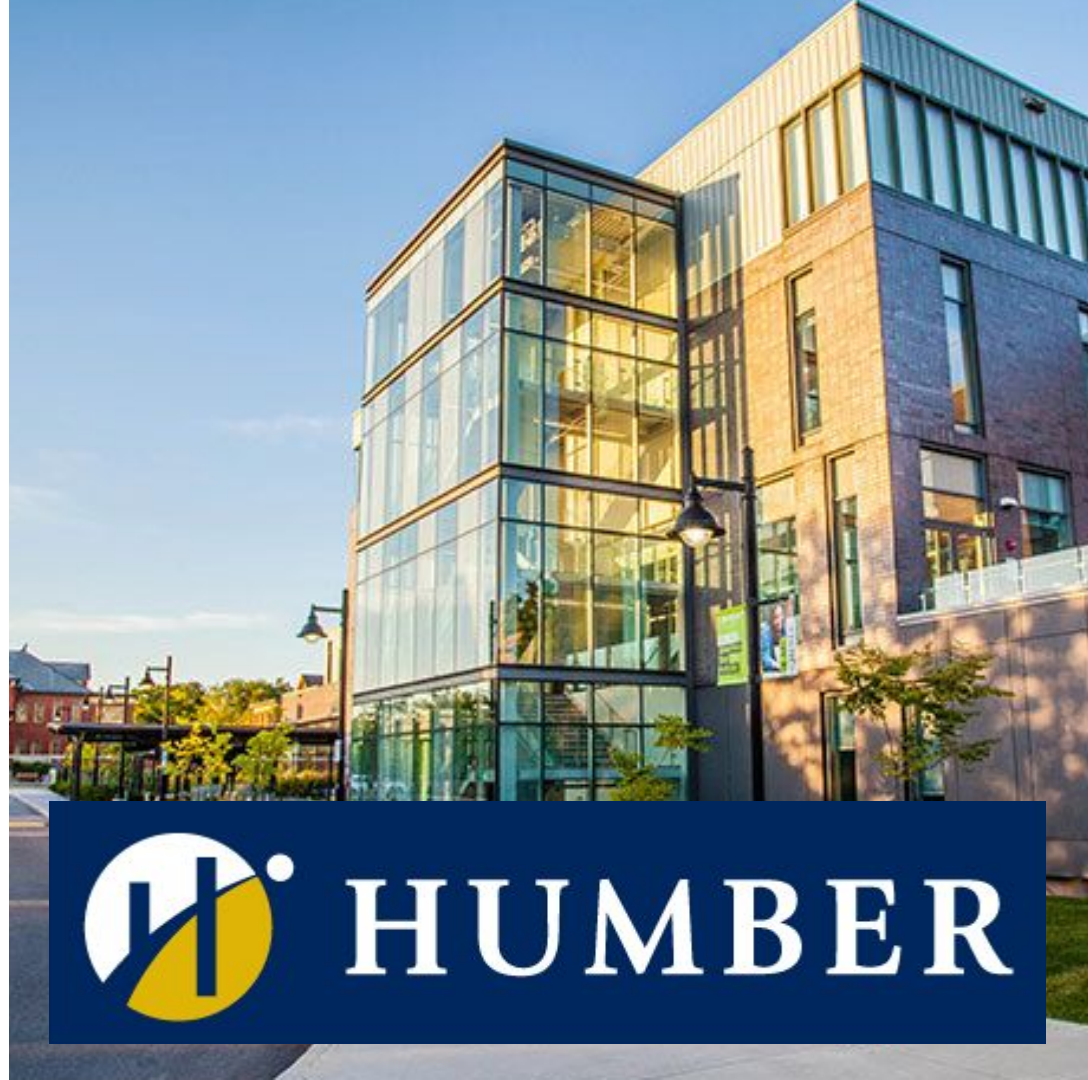
Product
(capital pi)

$$\prod_{n=1}^4 2n$$

```
prod = 1;  
for( n=1; n<=4; n++ )  
    prod *= 2*n;
```

HTTP-5114

JavaScript Review
APIs



HUMBER

URL Breakdown

Understanding a URL is crucial:

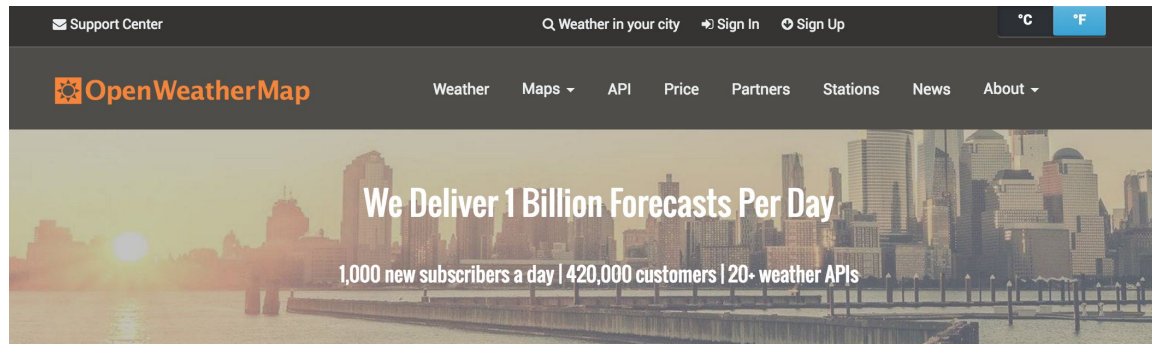
`http://www.domain.com/folder/file.html?variable=value&variable2=value2#anchor`

- Protocol: `http://`, `https://`, `ftp://`
- Domain: `www.domain.com`, `sub.domain.com`
- Folder: `/services/`
- File: `index.html`, `home.php`
- Variables: `?variable=value&variable2=value2`
- Anchor: `#home`, `#top`

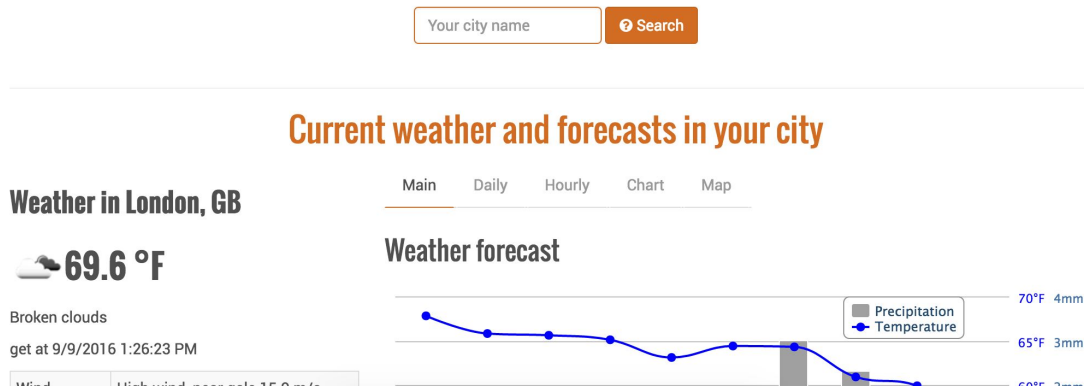
Note: `?` represents the end of the URL and the beginning of the variables

Note: `&` divides each individual variable

OpenWeatherMap



OpenWeatherMap is a free weather service with a RESTful API to allow developers to retrieve weather information.



OpenWeatherMap: Current Weather

What is the URL to request the current weather for Toronto?

- Endpoint: Current weather data
`http://api.openweathermap.org/data/2.5/weather`
- Format: XML
- Security: API Key
- Unit: Metric

OpenWeatherMap: Current Weather

- `http://api.openweathermap.org/data/2.5/weather`
- `q=Toronto,ca`
- `appid=69cd1126e263e700b1254e478aaa52af`
- `units=metric`
- `mode=xml`
- `callback=callbackFunction`

Note: Open link in a browser and review the data and load into an HTML file and locate JSONP differences..

OpenWeatherMap: Current Weather

Using the HTML file we started building in class retrieve the following information and display it on the page:

- City Name
- Temperature
- Icon
- Description

JSONP

(as in "JSON with Padding") is a method commonly used to bypass the cross-domain policies in web browsers (you are not allowed to make **AJAX** requests to a webpage perceived to be on a different server by the browser). JSON and JSONP behave differently on both the client and the server.

OpenWeatherMap: JSONP

<http://api.openweathermap.org/data/2.5/weather?callback=>